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RESEARCH

SBP-YOLO: a lightweight real-time model for detecting speed bumps and potholes toward intelligent vehicle suspension systems

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Abstract

Preview-based vehicle suspension systems enhance ride comfort and handling stability through proactive road anomaly detection and suspension control. To ensure adequate time for system response, higher vehicle speeds require detecting obstacles at greater distances, where they appear as smaller objects in the image space. This creates a critical challenge for embedded platforms: achieving high detection accuracy for distant small targets while maintaining real-time performance under computational constraints. To address this, we propose SBP-YOLO, a lightweight YOLOv11-based detector that integrates GhostConv and VoVGSCSPC for efficient multi-scale feature extraction. The model incorporates a P2 branch for small-object sensitivity and a Lightweight and Efficient Detection Head (LEDH) that reduces the computational overhead introduced by the P2 layer while maintaining accuracy. A hybrid training strategy employing normalized Wasserstein distance (NWD) loss, knowledge distillation, and albumentations augmentation enhances robustness to environmental variations. Experimental results demonstrate that SBP-YOLO achieves 87.0% mAP, exceeding YOLOv11n by 5.8%, with notable improvements in distant small-object detection. After TensorRT FP16 quantization, the model achieves 139.5 FPS on an NVIDIA Jetson AGX Xavier, representing a 12.5% speed improvement compared to the P2-enhanced YOLOv11, while simultaneously reducing energy consumption by 14.8%–174.84 mJ per frame. These findings validate SBP-YOLO as an efficient real-time solution for intelligent suspension systems.

Keywords Road anomaly detection · Lightweight model · Embedded vision · Intelligent suspension · YOLOv11

1 Introduction

Intelligent vehicle suspension systems have attracted attention in vehicle fields for their capability to simultaneously enhance ride comfort and handling stability. Modern implementations employ magnetorheological semi-active [1, 2] and fully active systems [3] that dynamically adjust damping and stiffness properties [4]. Road anomalies such as speed bumps and potholes significantly impact ride quality [5]. Conventional detection methods based on vehicle dynamics or multi-sensor fusion [6, 7] remain inherently reactive, providing limited vibration suppression through suspension deflection and acceleration measurements. Their fundamental limitation lies in the absence of preview capability, restricting them to post-disturbance response and consequently constraining performance potential [8]. The vision-based preview framework illustrated in Fig. 1 addresses these limitations through deep learning-based road irregularity anticipation. This approach not only offers more economical preview signals compared to LiDAR solutions, but also delivers richer semantic information, enabling more effective adaptive suspension control [9, 10].

While vision-based detection has demonstrated considerable advances, its practical deployment in automotive settings presents multiple critical challenges for embedded implementation:

- Lack of task-specific design: most models are not tailored for road anomaly detection, where targets often appear as small, low-texture objects in distant road regions, requiring specialized architectural adaptations [11, 12].
- Computational constraints: the limited processing power of embedded platforms restricts model complexity, forc-

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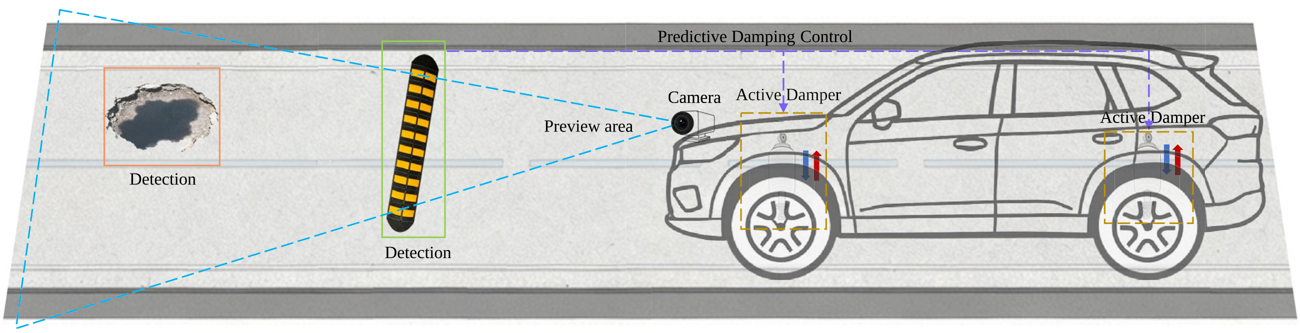


Fig. 1 Schematic of preview control principle with adjustable suspension for speed bumps and potholes

ing many lightweight models to sacrifice detection accuracy—particularly for small targets—in exchange for faster inference speed [13, 14].

- Environmental sensitivity: detection performance declines significantly under real-world driving conditions such as changing lighting, bad weather, and motion blur, indicating a lack of robustness in practical applications [15].

These challenges are particularly critical for suspension preview control, where system effectiveness is contingent upon reliable, early detection. As vehicle speed increases, the available preview time contracts substantially, imposing a non-negotiable demand for both low-latency inference and high detection accuracy. This stringent dual requirement exposes the inadequacy of existing approaches, which are hindered by a fundamental trade-off between these two objectives [16, 17]. This performance gap underscores the need for a new approach that can holistically address these dual requirements.

To bridge this gap, this paper introduces SBP-YOLO, a specialized framework that holistically optimizes accuracy, efficiency, and robustness for embedded road perception. The principal contributions are summarized as follows:

(1) Architecture and modules: A lightweight and efficient detection head (LEDH) is introduced to reduce the computational overhead of the P2 layer while preserving accuracy for small and distant targets. In combination with GhostConv modules and VoVGSCSPC blocks in the backbone and neck, the architecture improves feature extraction efficiency and robustness under diverse road and environmental conditions.

(2) Enhanced training methodology: A comprehensive training pipeline is employed to enhance performance in dynamic scenarios. It incorporates NWD loss for precise localization, BCKD knowledge distillation for effective feature transfer, and data augmentation with albumentations to simulate motion blur, illumination variations, and adverse weather.

(3) Hardware-aware optimization: Through FP16 quantization and TensorRT optimization, the model achieves 139.5 FPS on NVIDIA Jetson AGX Xavier. Hardware-level ablation studies and multi-power-mode analysis further demonstrate component contributions and energy efficiency, validating real-time deployment capability.

The remainder of this paper is organized as follows: Sect. 2 introduces the related work. Section 3 presents the proposed model architecture and the improved loss functions. Section 4 details the experimental setup and evaluation results. Section 5 discusses the model's inference performance on the embedded platform. Section 6 analyzes detection failure cases and suspension control applicability. Finally, conclusions are drawn in Sect. 7.

2 Related work

2.1 Deep learning object detection algorithms

Deep learning-based object detection has evolved through two primary paradigms. Two-stage detectors, exemplified by the R-CNN series [18–20], achieve high accuracy through sequential region processing but sacrifice computational efficiency, thereby limiting their real-time applicability. In contrast, single-stage detectors, represented by the YOLO family [21–29], perform direct bounding box regression in a single forward pass, enabling superior inference speed while maintaining competitive accuracy. Recent transformer-based detectors, including DETR [30] and RT-DETR [31], leverage self-attention for enhanced context modeling, yet face challenges in balancing accuracy with efficiency for embedded deployment. Existing lightweight YOLO variants [32, 33] remain constrained by limited small-target resolution, restricted computational budgets, and insufficient adaptation to road-specific environmental variations. Although the single-pass YOLO architecture is suitable for real-time perception, suspension preview control demands even lower latency and higher reliability, making targeted architectural refinements necessary.

2.2 Research status of speed bump and pothole detection

Road anomaly detection has advanced significantly with deep learning adoption, particularly for speed bumps and potholes. Early studies have identified illumination and weather variations as major challenges. Bučko et al. [15] reported clear performance degradation of YOLOv3 under low-light and rainy conditions, revealing limitations in complex environments.

Building upon these foundations, researchers have developed enhanced detection frameworks for road anomaly recognition. Park et al. [34] systematically compared several compact architectures and identified YOLOv4-tiny as achieving the optimal balance with 78.7% mAP@0.5. Edge deployment capabilities were further advanced by Asad et al. [11], who implemented Tiny-YOLOv4 on Raspberry Pi while maintaining 31.76 FPS at 90% accuracy. Subsequent studies introduced specialized architectures for improved road surface analysis. Wang et al. [35] proposed YOLOv5-FPNet with dynamic snake convolution and adaptive loss functions, reporting 38.76% mAP and 51.23% FPS improvements. Additional innovations include YOLOv8-based detectors [36] and the high-precision SBDNet [37]. Unified detection frameworks have also emerged, such as the lightweight dual-class network by Peralta-López et al. [38] and comprehensive comparisons between YOLO variants and SSD-MobileNet [39]. POT-YOLO [12] achieves 97.6% precision in pothole classification using an edge-enhanced YOLOv8 network, prioritizing detailed damage analysis over real-time control requirements. Recent surveys [40, 41] consistently identify environmental robustness as a critical challenge requiring further investigation.

Existing detection methods primarily target general perception tasks, neglecting the specialized demands of suspension control systems—particularly the need for cost-effective, high-precision detection amenable to embedded deployment. The interplay between detection performance and real-time suspension adjustment remains unexplored, while implementation aspects such as model optimization, quantization, and hardware-level energy efficiency receive limited attention. In response to these limitations, the SBP-YOLO framework is developed as a deployment-oriented solution that maintains detection accuracy while achieving computational efficiency tailored for suspension control applications.

2.3 Lightweight convolutions and detection components

Lightweight design remains a core focus in object detection research, with significant advances demonstrated across diverse domains [42–45]. Central to this progress is the optimization of convolutional architectures. Although standard convolutions (SC) form the basic building blocks of most detection networks, their high computational cost often restricts deployment under resource constraints. To mitigate this, GhostConv, introduced in GhostNet [46], employs efficient linear operations to significantly reduce parameters, while GSConv and VoVGSCSP [47] optimize the accuracy–efficiency balance in neck design.

These lightweight components underpin several recent embedded models. Slim-YOLOv8 [14] incorporates the slim-neck module from [47] to enhance feature extraction efficiency while maintaining accuracy. In parallel, Edge-YOLO [13] demonstrates that detection head optimization is equally crucial, employing an anchor-free structure and decoupled head design to achieve real-time performance on edge devices.

Existing research demonstrates that lightweight model design requires scenario-specific optimization to balance efficiency and accuracy. The SBP-YOLO framework addresses this need by integrating validated lightweight components with a novel LEDH, specifically designed to meet the challenges of small-target recognition and low-latency processing in automotive suspension systems.

3 Methods

3.1 SBP-YOLO architecture

This study focuses on intelligent suspension systems that require accurate and real-time detection of road surface anomalies, including potholes and speed bumps, to enable predictive control. To satisfy real-time constraints under limited computational resources, YOLOv11n is adopted as the baseline and redesigned into a lightweight and efficient framework named SBP-YOLO, as shown in Fig. 2. In the backbone, conventional convolutional layers at deeper stages are replaced with GhostConv modules, which reduce computational cost while preserving strong feature extraction capability. A novel VoVGSCSPC module is embedded in both the backbone and neck to enhance semantic feature propagation and multi-scale feature representation.

An additional P2 detection layer is introduced to enhance the perception of small and distant anomalies.

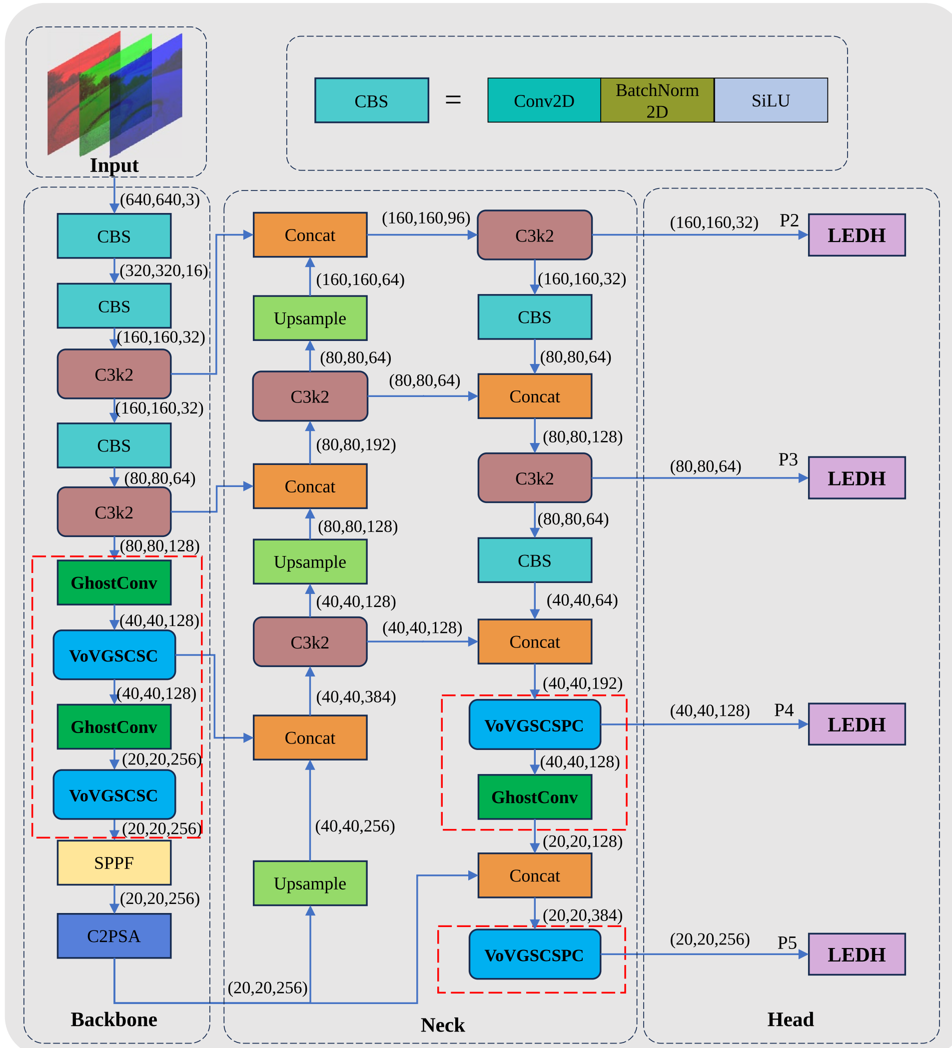


Fig. 2 SBP-YOLO framework with backbone, neck, and head. The neck fuses multi-scale features (P2–P5) from the backbone and passes them to the head for classification and regression

This modification extends the standard three-scale detection structure to a four-scale hierarchy composed of P2–P5 feature maps. The inclusion of the early P2 layer allows the network to capture finer spatial details and detect obstacles earlier, providing the suspension controller with more time to respond and adjust its dynamics. The detection head, referred to as LEDH, effectively balances computational efficiency with detection accuracy. All feature maps are annotated by their spatial dimensions, such as 320×320× 16, representing height, width, and channel depth for clarity of the multi-scale architecture.

3.2 GhostConv module

Efficient small-object detection requires a large receptive field with minimal computational overhead, which is difficult to achieve using SC. Stacking layers enlarges the receptive field but introduces redundancy and high FLOPs, while reducing layers sacrifices feature quality.

To balance this trade-off, the proposed model adopts the GhostConv operation from GhostNet [46] as a lightweight alternative to SC. As illustrated in Fig. 3a, GhostConv first extracts intrinsic feature maps via a standard convolution, then applies depthwise convolution on a subset of channels to generate additional "ghost" features. This two-stage process effectively enriches feature diversity while significantly reducing parameters and computation.

In Fig. 3b, let c_1 and c_2 denote the input and output channels, k the kernel size, and $H \times W$ the spatial resolution of the feature map. For SC, the parameter count and FLOPs are

$$P_{SC} = c_1 c_2 k^2, \quad F_{SC} = 2 c_1 c_2 H W k^2, \tag{1}$$

where P and F denote the number of parameters and the FLOPs, respectively, and $c_1, c_2, k, H \times W$ represent the input channels, output channels, kernel size, and spatial resolution of the feature map.

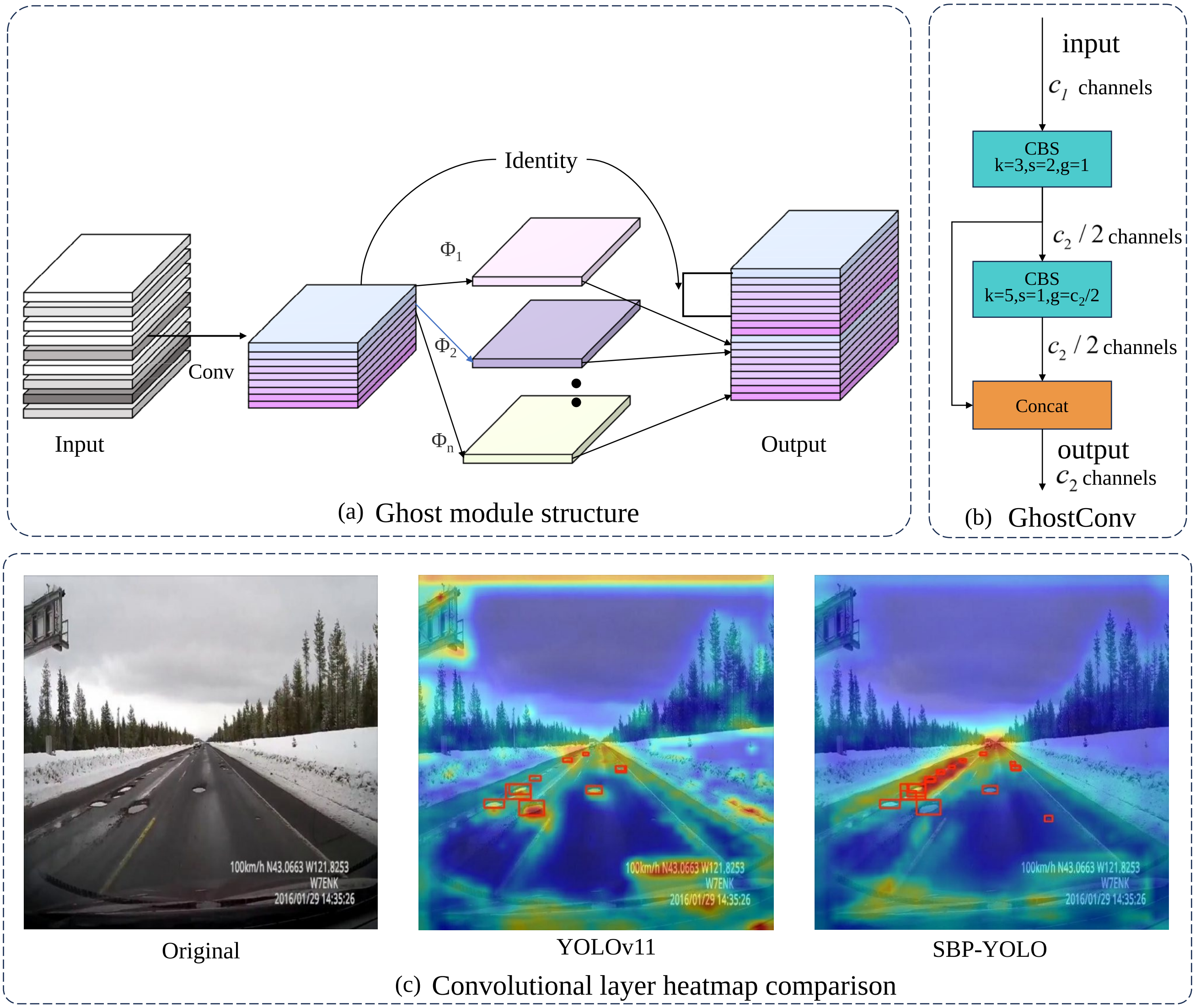


Fig. 3 Illustration of the Ghost module design and its impact on feature activation. **a** Ghost module; **b** GhostConv architecture; **c** comparison of backbone activation heat maps at the fifth and seventh convolutional layers

For GhostConv, the main convolution uses kernel size k_m and the depthwise convolution uses kernel size k_c. The corresponding parameters and FLOPs are

$$P_g = c_1 \cdot \frac{c_2}{2} \cdot k_m^2 + \frac{c_2}{2} \cdot k_c^2, \tag{2}$$

$$F_g = 2c_1 \cdot \frac{c_2}{2} \cdot HWk_m^2 + 2 \cdot \frac{c_2}{2} \cdot HWk_c^2, \tag{3}$$

with k_m, k_c denoting the kernel sizes in the GhostConv module.

The ratios relative to SC are

$$\frac{P_g}{P_{SC}} = \frac{k_m^2 + \frac{k_c^2}{c_1}}{2k^2}, \quad \frac{F_g}{F_{SC}} = \frac{k_m^2 + \frac{k_c^2}{c_1}}{k^2}. \tag{4}$$

Under typical settings ($c_1 \in [64, 256]$, $k_m = 3$, $k_c = 5$), GhostConv reduces parameters and FLOPs to 48–55% of SC, demonstrating its efficiency for embedded detection. Beyond computational savings, Fig. 3c shows that the augmented backbone enhances responses to distant small-scale anomalies, indicating an improved receptive field and richer feature representation.

3.3 VoVGSCSPC module

To further strengthen multi-scale representation while maintaining efficiency, the proposed network integrates the VoVGSCSPC module [47], which combines the advantages of GhostConv and channel shuffle mechanisms.

As illustrated in Fig. 4a, GSConv first applies SC for downsampling, followed by DWConv on the resulting features. The outputs are concatenated and shuffled to generate the final feature map:

$$F_{\mathrm{GSC}} = \mathrm{Shuffle}\left(\mathrm{Cat}(\alpha(X_{C_1})_{C_2/2}, \delta(\alpha(X_{C_1})_{C_2/2}))\right)_{C_2}, \quad (5)$$

where X_{C_1} is the input with C_1 channels, α denotes SC, δ represents DWConv, and F_{GSC} is the output.

Building on GSConv, the GSBottleneck module (Fig. 4b) introduces a residual cross-stage design that stacks two GSConv blocks to enhance feature reuse and multi-scale representation:

$$\mathrm{GSB}_{\mathrm{out}} = F_{\mathrm{GSC}}(F_{\mathrm{GSC}}(\alpha(X_{C_1})_{C_1/2})) + \delta(X)_{C_1/2}. \quad (6)$$

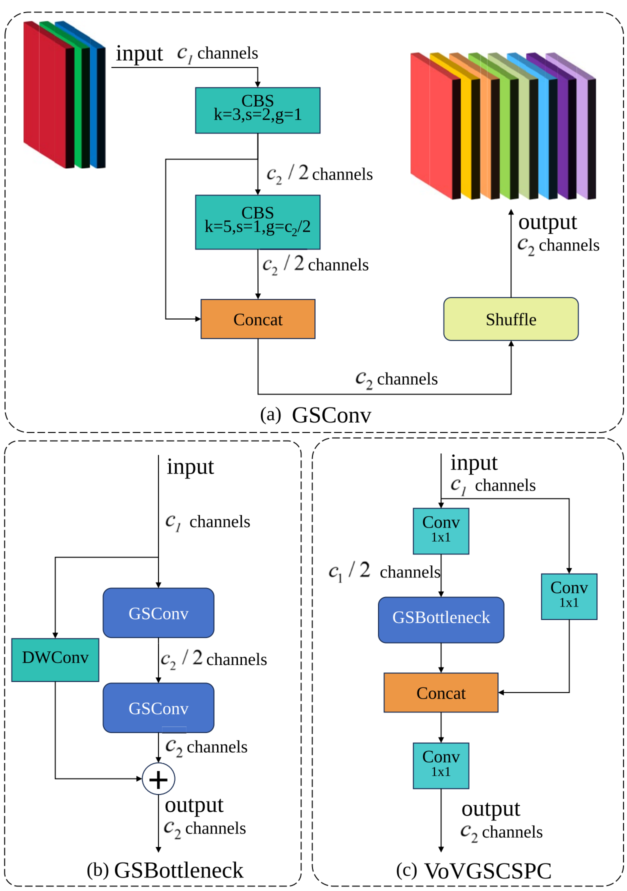


Fig. 4 Architectures of **a** GSConv, **b** GSBottleneck, and **c** VoVGSCSPC

Finally, the VoVGSCSPC module [47] (Fig. 4c) extends this design with a cross-stage partial architecture. The input is split into two branches: one passing through SC and GSBottleneck, and the other through a SC branch. Their outputs are fused as

$$\mathrm{VoVGSCSP}C_{\mathrm{out}} = \alpha\left(\mathrm{Concat}(\mathrm{GSB}_{\mathrm{out}}, \alpha(X_{C_1}))\right), \quad (7)$$

where $\mathrm{VoVGSCSP}C_{\mathrm{out}}$ denotes the final output.

VoVGSCSPC modules enhance multi-branch feature extraction in complex road environments through enriched representation learning and improved multi-scale fusion. This architecture significantly boosts detection performance for small-scale anomalies while maintaining real-time embedded efficiency.

3.4 Lightweight and efficient detection head (LEDH)

YOLOv11 improves small-object detection by introducing a shallow P2 layer, but its heads still use two consecutive 3×3 convolutions for classification and regression, resulting in high computational cost and limiting embedded deployment. To address this, we propose LEDH, which reduces redundancy while preserving accuracy for small targets by leveraging multi-scale features from the neck (Fig. 5a).

As illustrated in Fig. 5b, for each feature scale i, the input feature map c_i is first processed by two stacked grouped convolutions (GroupConv) with group count $g = c_i/16$:

$$\begin{aligned} F_i &= \mathrm{Conv}_g\left(\mathrm{Conv}_g(c_i)\right), \\ H_i &= \mathrm{Concat}\left[\mathrm{Conv}_{1\times1}^{4r}(F_i),\ \mathrm{Conv}_{1\times1}^{n_c}(F_i)\right]. \end{aligned} \quad (8)$$

The group count g was determined empirically to balance computational efficiency and inter-channel feature interaction, minimizing excessive cost while avoiding overly restricted communication. F_i denotes the shared transformed feature map, and H_i represents the final detection head output containing bounding box regression logits and class probabilities. n_c and r correspond to the number of classes and the distribution focal loss (DFL) bin count, respectively.

In contrast to YOLOv11's dual-branch design, LEDH employs a shared GroupConv transformation that allows both classification and regression branches to utilize enriched feature representations. This approach reduces computational redundancy while maintaining multi-scale representation capacity, significantly decreasing parameters and FLOPs. Lightweight 1×1 convolutions then generate predictions, yielding an efficient head for embedded deployment.

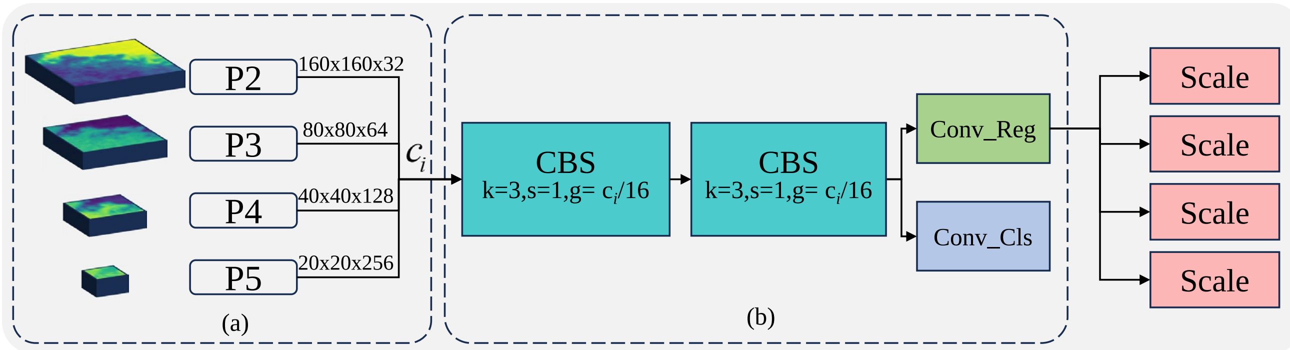


Fig. 5 Structure of the proposed LEDH. **a** Multi-scale feature maps (P2–P5) from the neck; **b** the proposed LEDH detection head

3.5 Hybrid loss training strategy

The limited pixel representation and blurred boundaries of small objects significantly amplify the sensitivity of IoU-based metrics to minor localization deviations. To address this issue, we introduce the normalized Wasserstein distance (NWD) [48] as a distribution-aware metric. This approach models bounding boxes as two-dimensional Gaussian distributions, providing smoother gradients during positional shifts and enhanced robustness to scale variations. By integrating NWD with conventional IoU loss, the model achieves substantial improvement in localization accuracy for small targets.

Given bounding boxes $A = (cx_a, cy_a, w_a, h_a)$ and $B = (cx_b, cy_b, w_b, h_b)$, the squared 2-Wasserstein distance between their corresponding Gaussian distributions $\mathcal{N}_a$ and $\mathcal{N}_b$ is computed. Since this distance does not directly serve as a similarity measure, it is normalized as:

$$\mathrm{NWD}(\mathcal{N}_a, \mathcal{N}_b) = \exp\left(-\frac{\sqrt{W_2^2(\mathcal{N}_a, \mathcal{N}_b)}}{C}\right), \tag{9}$$

where C is a dataset-specific scaling constant (e.g., 0.5) determined via ablation studies. The final bounding box regression loss is defined as:

$$\begin{cases} L_{\mathrm{NWD}} = 1 - \mathrm{NWD}(\mathcal{N}_a, \mathcal{N}_b) \\ L_{\mathrm{obj}} = (1-\alpha)\cdot L_{\mathrm{NWD}} + \alpha \cdot L_{\mathrm{iou}}, \end{cases} \tag{10}$$

where L_{NWD} represents the Wasserstein-based loss, L_{iou} is the standard IoU loss, and $\alpha = \mathrm{IoU}_{\mathrm{ratio}} \in [0, 1]$ controls the loss weighting.

4 Experiments and analysis

4.1 Experimental settings

The experiments were conducted on Ubuntu 24.04 with an Intel® Xeon E5-2680 CPU (2.4 GHz) and an NVIDIA Tesla V100 GPU (16 GB). The software environment comprised PyTorch 1.13.1 and CUDA 11.7. The model was trained using the Adam optimizer with an initial learning rate of 0.001, batch size of 16, and 300 epochs. Input images were resized to 640×640 to enhance convergence and computational efficiency. Early stopping was employed, terminating training if validation performance did not improve for 30 consecutive epochs.

4.2 Dataset construction

Existing road surface datasets provide samples of potholes and speed bumps but often lack sufficient variation in illumination, weather, surface materials, and geographic conditions. To address these limitations, we constructed a composite dataset by aggregating, cleaning, filtering, and re-annotating samples from multiple public sources, including benchmark datasets [15, 38, 49, 50] and community platforms such as Kaggle and Roboflow Universe.

The dataset construction applied strict filtering criteria to ensure data quality. For potholes, minor cracks, shallow depressions, and weak anomalies were excluded, retaining only geometrically significant defects; this subset includes 565 texture-focused images to enhance fine-grained surface

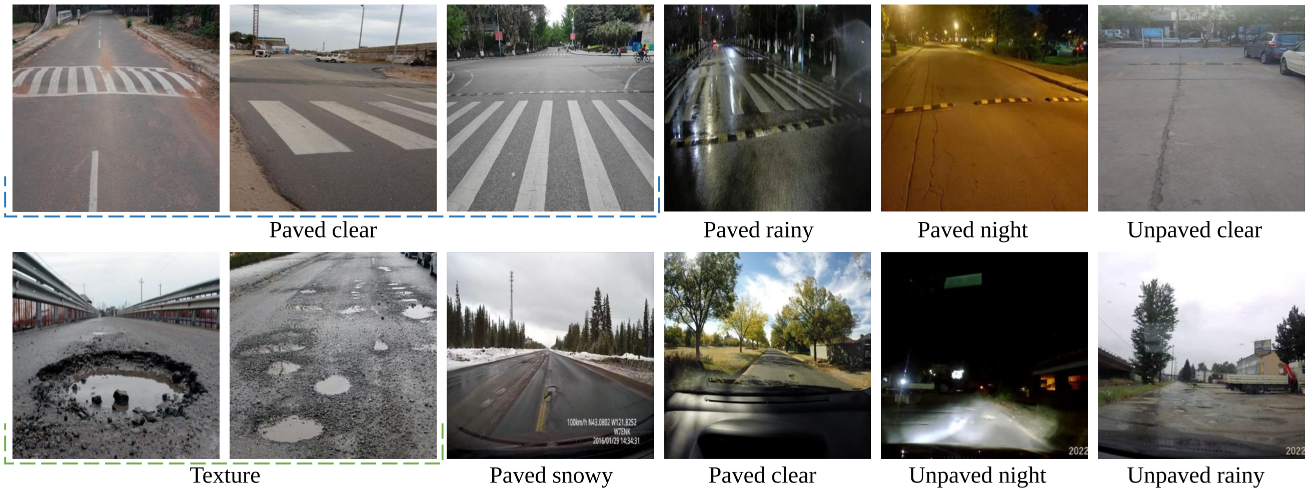


Fig. 6 Original images in the dataset, including potholes and speed bumps under diverse weather and lighting conditions

pattern representation, and each image may contain multiple instances to reflect realistic road conditions. For speed bumps, only clearly defined and drivable structures were retained, with ambiguous or irregular objects excluded. After this filtering process, the final dataset contains 5500 images, comprising 3000 pothole images and 2500 speed bump images. Example images are shown in Fig. 6.

The dataset comprehensively covers diverse environmental conditions, with its detailed composition and statistical distribution summarized in Table 1. The pothole subset is well represented across both paved and unpaved roads under varying illumination and weather conditions, while speed bump images, though predominantly captured in clear scenarios, include meaningful samples from nighttime and rainy environments. This balanced yet varied distribution supports robust cross-domain generalization under changes in illumination, road surface type, and weather conditions.

Real-world adverse-weather images are limited. To improve model robustness, data augmentation was applied using albumentations [51], including geometric transformations (flipping, cropping, rotation), photometric adjustments (brightness, contrast, hue, saturation), and synthetic weather effects (rain, fog, snow), illustrated in Fig. 7. Pothole images were increased by 50% to account for the high variability in instance number, arrangement, and surface texture, whereas speed bump images were increased by 20% because

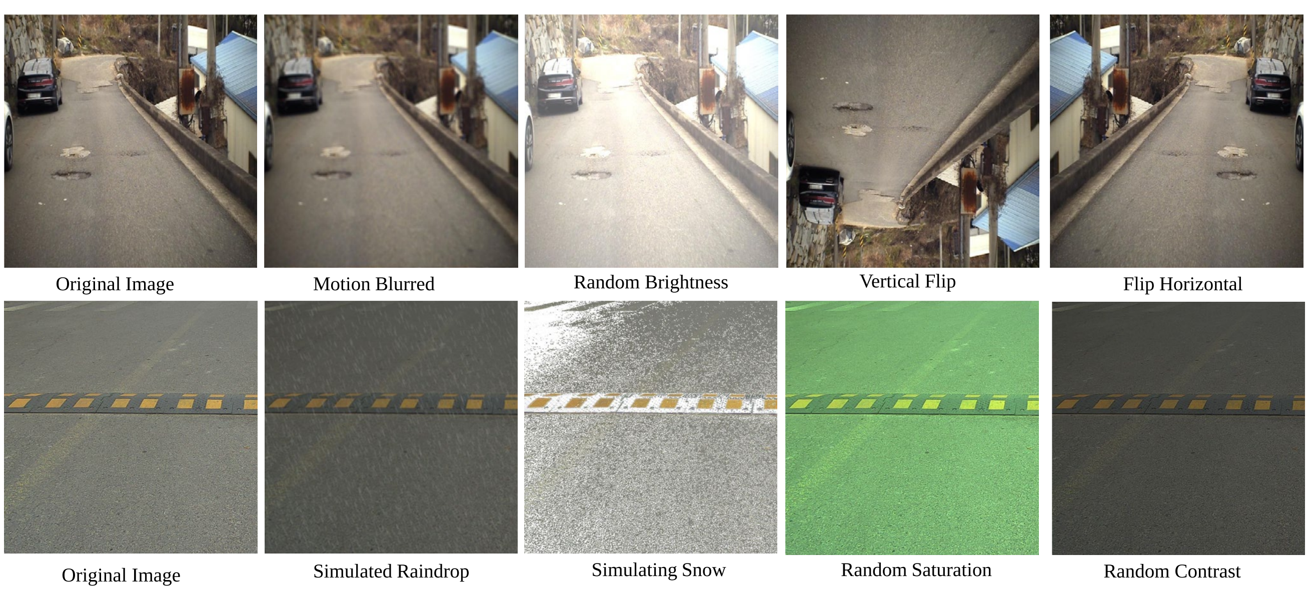


Fig. 7 Comparison of the effects after some image data augmentation

Table 1 Composition and statistical distribution of the original dataset

Category	Sub-category	Condition	Images	Instances
Potholes (3000 images)	Texture	Clear	565	2062
	Paved	Clear	1835	4599
	Paved	Snowy	132	430
	Unpaved	Dusk	243	565
	Unpaved	Night	108	174
	Unpaved	Rainy	117	325
Speed bumps (2500 images)	Paved	Clear	1982	2244
	Paved	Night	36	38
	Paved	Rainy	31	31
	Unpaved	Clear	451	458

each image typically contains a single, relatively uniform instance. The final augmented dataset contains 7500 images.

The augmented dataset was randomly divided into 5250 training, 1125 validation, and 1125 test images (7:1.5:1.5), maintaining class balance across splits. Random stratified partitioning preserves diversity in illumination, road type, and weather conditions. The dataset provides a reproducible and comprehensive foundation for training and evaluating vision-based suspension perception models under diverse operational scenarios.

4.3 Evaluation indicators

Detection performance is evaluated using mean average precision (mAP), including mAP_{50} and the average over IoU thresholds from 50 to 95% (mAP_{50-95}). Model efficiency is assessed via parameters (Params), GFLOPs, network depth (Layers), FPS, and inference latency. FPS and latency are measured with batch size = 1, reflecting typical embedded deployment where images are processed sequentially; batch = 1 performance directly determines real-time capability, as large-batch optimizations are inapplicable. Layers approximate architectural and spatial complexity: deeper networks may improve representational capacity, but increase inference time and memory footprint. For a single class, precision (P) and recall (R) are

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \tag{11}$$

where TP, FP, FN denote true positives, false positives, and false negatives. Average precision (AP) and mean average precision (mAP) are

$$\mathrm{AP} = \int_0^1 P(R)\,\mathrm{d}R, \quad \mathrm{mAP} = \frac{1}{M}\sum_{i=1}^{M} \mathrm{AP}_i, \tag{12}$$

with M object categories.

4.4 Ablation experiment

4.4.1 Impact of model components

Ablation studies in Table 2 assess the contribution of each module. The baseline YOLOv11n achieves fast inference (71.2 FPS), but limited accuracy (mAP_{50} = 81.2%, mAP_{50-95} = 47.7%) due to missing low-level features for small objects, with a relatively shallow network. Adding a P2 detection head increases accuracy (mAP_{50} = 83.9%), but adds layers and computational cost, reducing speed. Replacing the P2 head with LEDH improves precision and recall while reducing FLOPs, demonstrating that architectural enhancements can increase representational power without excessive overhead. GhostConv and VoVGSCSPC consistently improve detection, with VoVGSCSPC reducing complexity and GhostConv minimizing FLOPs. Incorporating NWD loss further enhances localization of small and irregular anomalies, achieving the best overall performance (mAP_{50} = 86.6%, mAP_{50-95} = 51.1%) with

Table 2 Ablation study of different modules in SBP-YOLO

P2	P2-LEDH	Ghost	VoVG	NWD	P	R	mAP_{50}	mAP_{50-95}	GFLOPs	Params	FPS	Layers
					85.0	75.0	81.2	47.7	6.3	2.58	**71.2**	**238**
		✓			85.7	76.3	82.8	49.2	5.9	2.29	70.8	247
			✓		85.5	75.4	82.7	49.2	5.8	2.54	67.6	288
		✓	✓		85.4	76.4	83.1	49.6	**5.4**	2.25	65.9	297
		✓	✓	✓	86.2	76.2	82.7	49.1	**5.4**	2.25	65.9	297
✓					84.8	76.5	83.9	49.8	10.3	2.67	53.5	283
	✓				86.6	77.6	84.5	50.1	6.7	2.37	69.6	240
	✓	✓			86.3	78.1	84.5	49.8	6.2	2.09	67.3	249
	✓	✓	✓		87.5	79.2	84.6	49.9	5.8	2.04	63.7	299
	✓	✓	✓	✓	**89.0**	**79.4**	**86.6**	**51.1**	5.8	**2.04**	63.7	299

P2: YOLOv11 head on P2; P2-LEDH: proposed LEDH head on P2; Ghost: GhostConv; VoVG: VoVGSCSPC; NWD: introducing NWD loss. P, R, mAP_{50}, and mAP_{50-95} are in %; Params in M

Bold indicate the best results among the compared methods/models in thecorresponding column

Table 3 Sensitivity analysis of the LEDH grouping factor (group = c_i/n)

Grouping (n)	P (%)	R (%)	mAP_{50} (%)	mAP_{50-95} (%)	GFLOPs	Params (M)
$n=1$	85.7	79.5	84.8	50.3	4.9	1.91
$n=2$	84.9	77.5	83.8	49.0	5.0	1.92
$n=4$	88.7	77.6	85.3	50.3	5.1	1.94
$n=8$	85.9	77.8	83.6	49.1	5.3	1.97
$n=16$	87.5	79.2	84.6	49.9	5.8	2.04
$n=32$	86.1	79.9	85.2	51.3	6.7	2.18
$n=c_i$	87.8	77.4	85.3	49.8	8.7	3.47

Precision (P), recall (R), mAP_{50}, and mAP_{50-95} are reported in %; Params in M

moderate network depth. These results confirm that each module contributes to both accuracy and efficiency, and that network depth serves as an informative proxy for the model's architectural and spatial complexity in resource-constrained embedded deployment.

4.4.2 Effect of LEDH grouping factor

To evaluate the effect of channel grouping on LEDH, we vary the grouping factor n (groups = c_i/n). As shown in Table 3, excessive grouping reduction ($n=32$) achieves the highest recall but incurs higher computational cost, whereas small grouping ($n=4$) yields the highest precision but relatively low recall. A moderate setting ($n=16$) provides a balanced trade-off, achieving a recall of 79.2% and competitive precision while keeping computation efficient. This demonstrates that $c_i/16$ offers an optimal balance between detection reliability and real-time efficiency for suspension preview control.

4.5 Knowledge distillation experiment

Knowledge distillation (KD) transfers knowledge from a larger, well-trained teacher model to a lightweight student model, aiming to enhance generalization with minimal computational overhead. The proposed KD framework adopts a hybrid strategy, integrating logit-level supervision using BCKD [52] and feature-level guidance via CWD [53]. Six corresponding intermediate layers are selected from both models to enable effective feature alignment. The detailed training hyperparameters are summarized in Table 4. In our study, a teacher model with similar architecture and S-level scale (20.6 GFLOPs, 7.42M Params) was employed to guide the training of the student model. As shown in Table 5, KD leads to consistent improvements across all metrics. In particular, mAP_{50-95} increased by 3.5% on the validation set and 3.4% on the test set, indicating not only enhanced detection performance but also improved generalization. The comparable gains across both sets suggest that the student model benefits from the teacher's guidance without exhibiting overfitting.

Table 4 Hyperparameter settings for the proposed knowledge distillation (KD) framework

Hyperparameter	Value
Training epochs	500
Batch size	32
Optimizer	SGD
Logits distillation loss	BCKD
Feature distillation loss	CWD
Distillation loss schedule	Constant
Logits loss weight	0.5
Feature loss weight	0.5
Teacher layers used	12, 15, 18, 21, 24, 27
Student layers matched	12, 15, 18, 21, 24, 27

Table 5 Performance comparison of distilled model, teacher model, and non-distilled baseline on validation and test sets

	P (%)	R (%)	mAP_{50}	mAP_{50-95}
Validation set				
Baseline (student)	89.0	79.4	86.6	51.1
+ distillation	90.2	80.7	87.0	54.6
Teacher model	91.0	81.4	87.4	53.8
Test set				
Baseline (student)	88.6	78.2	86.3	51.5
+ distillation	89.9	81.8	87.7	54.9
Teacher model	89.9	82.0	87.8	54.1

4.6 Experimental comparison and analysis

4.6.1 Model comparison experiment

To comprehensively evaluate detection accuracy and computational efficiency, SBP-YOLO is benchmarked against representative detectors, including YOLOv3–YOLOv13, transformer-based RT-DETR, and multiple YOLOv11 variants with different scales and P2-enhanced configurations. This broad comparison ensures a fair assessment across architectures and computational budgets.

As shown in Table 6, SBP-YOLO consistently achieves a superior trade-off between detection accuracy and computational efficiency. Increasing the model size in the YOLOv11 family improves mAP_{50}, but the gains diminish as the model grows. YOLOv11n, YOLOv11s, and YOLOv11m achieve mAP_{50} scores ranging from 81.2 to 83.7% while utilizing 2.58 M–19.10 M parameters. YOLOv11l reaches 83.9% mAP_{50} with 24.11 M parameters. Variants with the P2 detection head improve accuracy slightly, but require much more computation. YOLOv11-P2-s attains 86.3% mAP_{50} with 9.16 M parameters and 29.2 GFLOPs, comparable to SBP-YOLO while being nearly five times heavier.

In contrast, SBP-YOLO achieves 86.6% mAP_{50} with only 2.04 M parameters and 5.8 GFLOPs, outperforming YOLOv11n-P2, Hyper-YOLO-t (83.6%), Mamba-YOLO-t (82.3%), and RT-DETR-l (81.3%) with far lower computation. Knowledge distillation further boosts SBP-YOLO to 87.0% mAP_{50} and 54.6% mAP_{50-95} without additional cost, demonstrating its ability to enhance feature representation efficiently.

Training curves in Fig. 8 show that SBP-YOLO achieves slightly higher mAP_{50} than YOLOv11 baselines and P2-enhanced variants, including a modest advantage over YOLOv11s-P2 (Fig. 8a). The bounding-box loss converges faster and more stably for SBP-YOLO compared to other models (Fig. 8b), indicating that its lightweight design not only maintains competitive accuracy but also improves training stability.

4.6.2 Multi-environment performance evaluation

As shown in Fig. 9a, SBP-YOLO exhibits improved detection capability for small-scale road features at extended distances, such as far-field speed bumps and potholes. These targets often appear with reduced resolution and weak texture cues, posing challenges to existing detection models. Compared to baseline YOLO variants, SBP-YOLO achieves higher accuracy and greater consistency in identifying such long-range and low-visibility objects. Figure 9b illustrates detection performance under varied illumination conditions, including sunset, nighttime, and intense sunlight. SBP-YOLO maintains stable accuracy across all scenarios. At night, it outperforms YOLOv5 by 9%, and during sunset—where reduced contrast can hinder detection—SBP-YOLO shows more robust results than other models. Under strong sunlight that induces glare, SBP-YOLO successfully identifies distant potholes, whereas models like YOLOv8 occasionally miss the target.

To evaluate robustness in visually degraded environments, Fig. 9c reports performance on synthetic datasets with rain, snow, and motion blur. SBP-YOLO attains 0.64 detection accuracy in rainy scenes, exceeding YOLOv11n (0.47) and YOLOv5n. Consistent gains are observed under snow and blur, confirming its resilience to diverse visual perturbations. Figure 9d further validates performance in real adverse weather. For rainy speed bump detection, SBP-YOLO achieves 0.72 confidence, outperforming YOLOv11 (0.59) and YOLOv8 (0.64). In snowy pothole detection, it maintains 0.6 confidence where YOLOv5/v11 miss nearby targets and YOLOv8 scores only 0.28, demonstrating reliable detection in challenging real-world conditions.

These comprehensive experiments demonstrate that SBP-YOLO provides a robust visual perception solution for intelligent suspension systems, capable of maintaining reliable detection performance across varying distances, illumination conditions, and adverse weather scenarios.

4.6.3 Evaluation of different detection heads

To evaluate LEDH, we replaced only the detection head in SBP-YOLO while maintaining the original backbone and neck architecture. As summarized in Table 7, all compared detection heads are designed with lightweight and efficiency-oriented architectures. The original YOLOv11 head achieves 84.8% mAP50 at 51.2 FPS, but requires a relatively deep network structure. While both LADH and LSCD improve accuracy, this comes at the cost of increased or inconsistent network depth; LSCSBD shows reduced detection performance, and LSDECD introduces higher computational complexity. In contrast, the proposed LEDH achieves 84.6% mAP50 and 79.2% recall with the lowest computational cost (5.8 GFLOPs), moderate parameter count (2.04M), highest inference speed (63.7 FPS), and well-controlled network depth (299 layers). These results collectively demonstrate that LEDH establishes an optimal balance between accuracy, efficiency, and architectural complexity, rendering it highly suitable for real-time embedded deployment.

5 Experiment in embedded device

5.1 Experimental environment

All embedded inference experiments were performed on an NVIDIA Jetson AGX Xavier platform equipped with a Volta GPU (512 CUDA cores, 64 Tensor Cores) and 16 GB LPDDR4 memory. The device supports up to 32 TOPS at INT8 precision within a configurable 10–30 W power envelope. To evaluate on-device robustness, SBP-YOLO was tested under four power configurations: 10 W, 15 W, 30 W, and the maximum-performance mode. In the maximum-performance setting, the cooling fan was fixed at full speed to prevent thermal throttling and ensure stable peak-frequency operation.

As shown in Fig. 10, the test environment consists of the Xavier module and its carrier board. Inference was executed

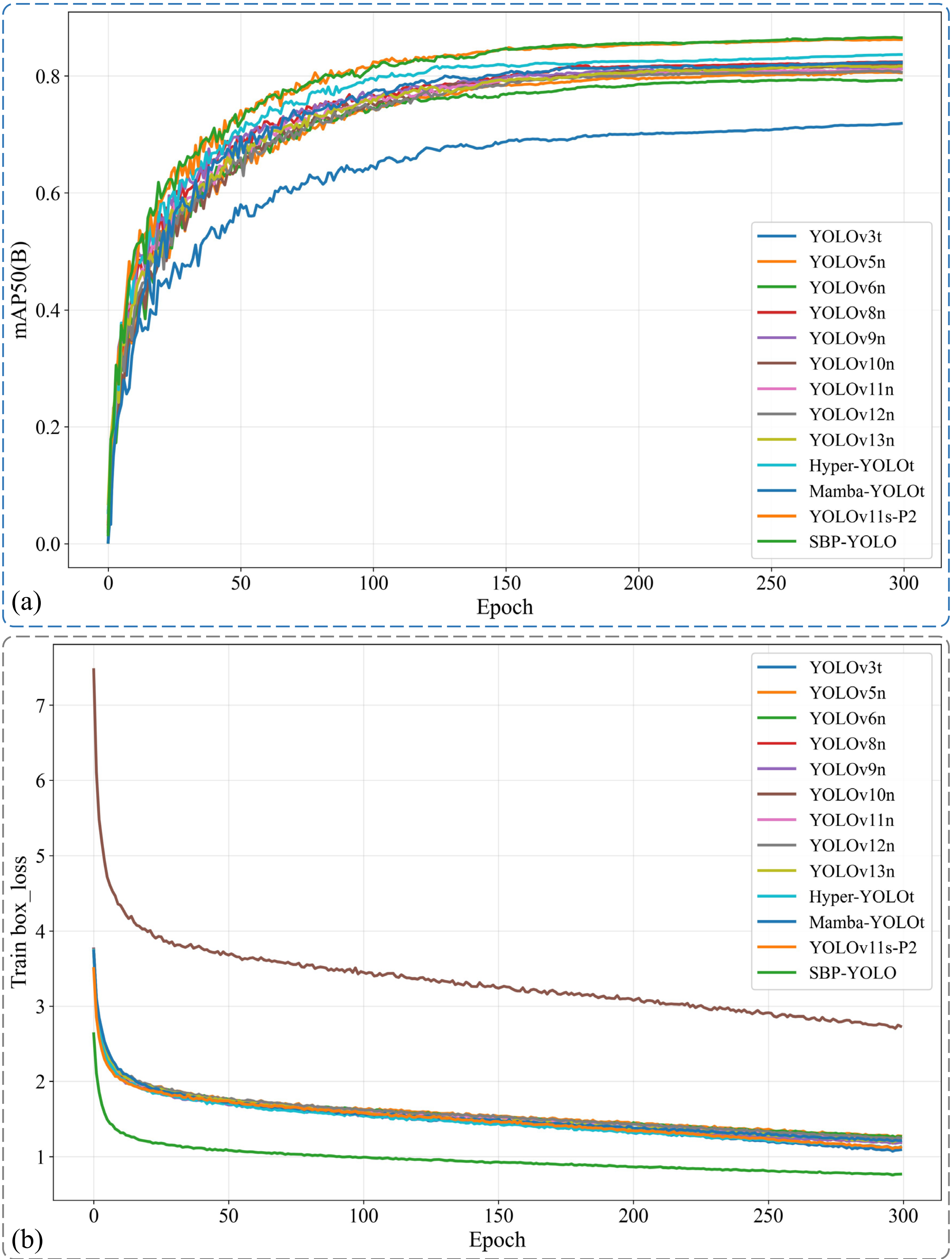


Fig. 8 Training curves over 300 epochs for different YOLO models: **a** mAP_{50}, **b** bounding box loss

via command line with real-time throughput measurement. System status was continuously monitored using `tegrastats`, which reports GPU utilization, memory-controller load, temperature, and instantaneous power consumption, enabling detailed assessment of latency stability across different power modes.

The SBP-YOLO model was converted and deployed using TensorRT, with optimization supported by an open-source YOLO deployment tool [58]. All experiments were conducted using NVIDIA JetPack 5.1.5.

5.2 Hardware-level ablation study

To assess the impact of individual architectural modules on embedded performance, we conducted a hardware-level ablation study on the NVIDIA Jetson AGX Xavier at maximum power. Each configuration processed 100 images repeated 100 times to minimize statistical variance, with system power continuously monitored. Reported metrics (FPS, latency, power, FPS/W) represent averaged values for reliable estimation.

Table 8 summarizes the results across module combinations and precisions. Adding the P2 detection head increases computational demand, reducing FPS from 229.83 to 152.78 and lowering efficiency from 11.22 to 6.54 FPS/W (INT8). Incorporating LEDH and Ghost modules partially restores throughput (up to 168.31 FPS) and improves efficiency (up to 7.17 FPS/W), reflecting more efficient feature computation without expanding model capacity. Hardware utilization analysis reveals that GPU usage increases with model complexity (91.19–93.49% under INT8), while RAM consumption remains relatively stable across configurations.

Within equivalent architectural configurations, SBP-YOLO provides consistent gains in both inference speed and energy utilization. Under FP16 precision, SBP-YOLO achieves 139.50 FPS, representing 12.5% higher speed and 17.5% better efficiency compared to YOLOv11-P2 at similar power consumption, while reducing energy per frame by 14.8% (from 205.27 mJ to 174.84 mJ). Under FP32 precision, these advantages expand to 19.8% speed improvement and 24.5% efficiency gain, accompanied by a 19.7% reduction in energy per frame (from 408.19 mJ to 327.97 mJ). These results demonstrate that despite the computational overhead of the P2 head, the proposed architectural refinements maintain high computational efficiency while better utilizing constrained embedded resources.

5.3 Power-mode effects on energy efficiency and latency

In vehicular operational environments, energy consumption analysis of embedded perception systems is critical, as both power utilization and latency variations directly impact real-time reliability. To characterize these effects, we evaluate multiple YOLO architectures across various power modes. As illustrated in Fig. 11, under identical 15W power constraints, SBP-YOLO consistently demonstrates superior energy efficiency compared to other YOLO variants. With INT8 quantization, it achieves 4.9 and 48.0% reductions in energy per frame relative to YOLO11n-P2 and YOLO11s-P2, respectively. The advantage is further amplified with FP16 quantization, yielding 16.4 and 55.6% lower energy consumption. Under FP32 precision, SBP-YOLO maintains this competitive edge with 20.0 and 65.7% energy savings, demonstrating its exceptional capability to leverage embedded power constraints efficiently.

FP16 precision presents an optimal balance for practical deployment considerations. SBP-YOLO configured with FP16 at 15W maintains competitive energy efficiency (89.7 mJ/frame) while preserving enhanced numerical precision, rendering it particularly suitable for applications demanding improved detection accuracy.

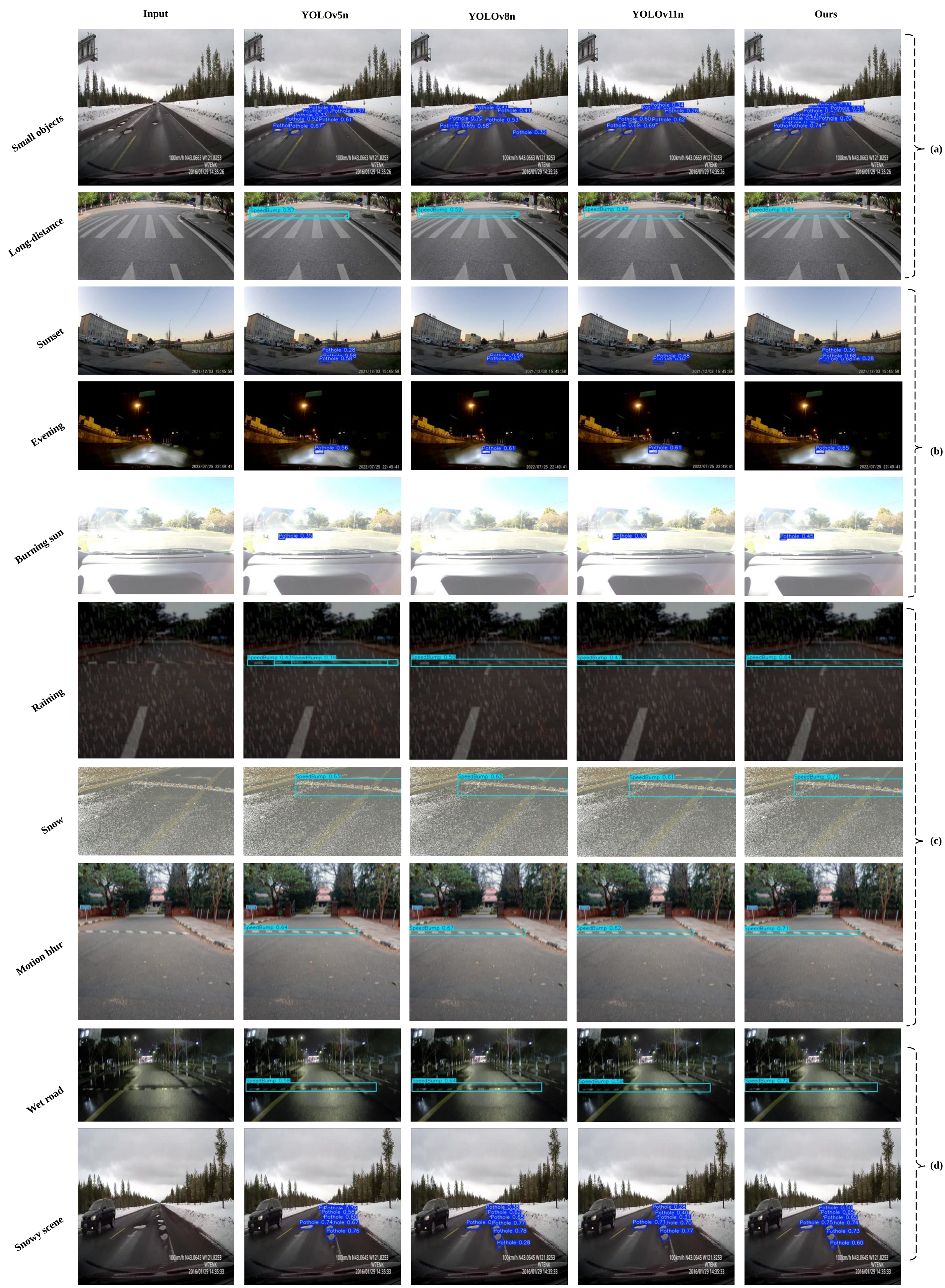
Empirical results confirm that 15W achieves the best energy efficiency, while higher power modes show diminishing returns and greater latency fluctuations. This highlights the need to co-optimize model design, quantization, and power management for reliable automotive deployment.

6 Discussion

6.1 Discussion on system failure detection

As illustrated in Fig. 12, different TensorRT quantization strategies exhibit distinct failure behaviors. INT8 quantization introduces noticeable degradation on small or distant targets, leading to missed detections in scenarios such as far-range speed bumps and multi-pothole road surfaces. These failure cases reflect the reduced representational precision of INT8 when handling low-texture or small-scale features. In contrast, TensorRT-FP16 maintains detection performance comparable to TensorRT-FP32 and the original PyTorch model, showing no significant degradation across the evaluated scenes. These observations highlight the need to account for quantization-induced uncertainty when deploying perception models on embedded systems. Specifically, INT8 quantization should be used cautiously in scenarios requiring reliable long-range or small-object recognition, whereas FP16 provides a more robust accuracy–efficiency balance for safety-critical applications.

To further analyze model robustness, we conducted comprehensive inference on the entire dataset using SBP-YOLO and identified systematic failure patterns, as exemplified in Fig. 13. The primary failure modes include: object occlusion (Fig. 13a), where partial visibility prevents accurate contour recognition; distant low-texture


Input
YOLOv5n
YOLOv8n
YOLOv11n
Ours
Small objects
Long-distance
(a)
Sunset
Evening
Burning sun
(b)
Raining
Snow
Motion blur
(c)
Wet road
Snowy scene
(d)

◂**Fig. 9** Qualitative comparison of SBP-YOLO and baseline models under challenging conditions: **a** detection of small and distant features; **b** robustness to diverse illumination; **c** performance under simulated adverse weather and motion effects; **d** real-world adverse-weather scenarios, including wet road conditions and snowy environments with clear road surfaces

targets (Fig. 13b), where insufficient features hinder distant object detection; rain-affected roads (Fig. 13c), where water reflections mask road defect characteristics; and camera overexposure (Fig. 13d), where light saturation eliminates critical texture information. These scenarios consistently challenge the model's detection capability, emphasizing the importance of addressing environmental variability in real-world automotive applications.

6.2 Relation between perception latency and suspension preview control

The feasibility of vision-based suspension preview control is governed by the available preview time:

$$T_{\text{preview}} = D/v, \tag{13}$$

where D denotes the preview distance and v represents the vehicle speed. The total system latency must satisfy:

$$t_{\text{proc}} + t_{\text{ctrl}} + t_{\text{act}} + \Delta \le T_{\text{preview}}, \tag{14}$$

encompassing perception latency (t_{proc} = 7.17 ms), controller computation (t_{ctrl} = 50 ms), actuator response (t_{act} = 20 ms), and safety margin (Δ). The total latency budget is approximately $t_{\text{total}} \approx 77.17$ ms.

Table 6 Comparison of SBP-YOLO with representative YOLO variants, different-scale YOLOv11 models, and RT-DETR in terms of accuracy and computational efficiency

Models	P (%)	R (%)	mAP$_{50}$(%)	mAP$_{50-95}$(%)	Params (M)	GFLOPs	Size (MB)
YOLOv3t [21]	78.0	65.2	71.9	38.6	9.08	14.3	18.30
YOLOv5n [22]	83.4	74.5	80.6	46.1	2.18	5.8	4.43
YOLOv6n [23]	83.8	72.0	79.3	45.9	4.16	11.5	8.15
YOLOv8n [24]	88.2	73.5	82.4	48.3	2.68	6.8	5.36
YOLOv9t [25]	84.7	75.3	81.5	48.3	**1.76**	6.6	**3.95**
YOLOv10n [26]	84.8	72.5	82.0	48.3	2.27	6.5	5.48
YOLOv11n [27]	85.0	75.0	81.2	47.7	2.58	6.3	5.21
YOLOv11s [27]	88.0	75.1	83.7	50.0	8.98	21.3	18.28
YOLOv11m [27]	88.0	77.2	83.4	50.1	19.10	67.7	38.63
YOLOv11l [27]	86.3	76.6	83.9	50.5	24.11	86.6	48.80
YOLOv11n-P2	84.8	76.5	83.9	49.8	2.67	10.3	5.50
YOLOv11s-P2	87.9	79.1	86.3	51.5	9.16	29.2	18.77
YOLOv11m-P2	88.0	77.9	84.7	51.5	19.75	89.8	40.06
YOLOv11l-P2	86.8	76.8	83.8	49.9	25.00	114.1	50.78
YOLOv12n [28]	84.1	74.2	80.9	47.5	2.56	6.3	5.25
YOLOv13n [29]	84.4	75.0	81.6	48.2	2.19	5.9	4.77
Hyper-YOLO-t [32]	84.3	78.2	83.6	49.7	2.56	7.7	5.40
Mamba-YOLO-t [33]	84.4	76.2	82.3	48.9	5.40	12.3	11.13
RT-DETR-l [31]	84.5	75.6	81.3	39.0	30.51	103.4	63.15
SBP-YOLO (ours)	89.0	79.4	86.6	51.1	2.04	**5.8**	4.28
SBP-YOLO (distilled)	**90.2**	**80.7**	**87.0**	**54.6**	2.04	**5.8**	4.35

Bold indicate the best results among the compared methods/models in thecorresponding column

Table 7 Comparison of different detection heads on the SBP-YOLO framework. All models use the same backbone and neck; only the detection head differs

Model	P (%)	R (%)	mAP$_{50}$ (%)	mAP$_{50-95}$ (%)	GFLOPs	Params (M)	FPS	Layers
YOLOv11-head	86.7	76.8	84.8	50.0	9.4	2.34	51.2	342
LADH [54]	84.1	78.9	84.2	49.4	6.3	2.00	52.8	383
LSCD [55]	87.6	78.9	85.3	49.8	7.0	2.16	60.3	300
LSCSBD [56]	82.3	75.2	81.0	47.0	9.6	2.19	57.8	310
LSDECD [57]	86.3	77.5	84.2	49.5	10.2	1.98	56.0	301
LEDH (ours)	87.5	79.2	84.6	49.9	5.8	2.04	63.7	299

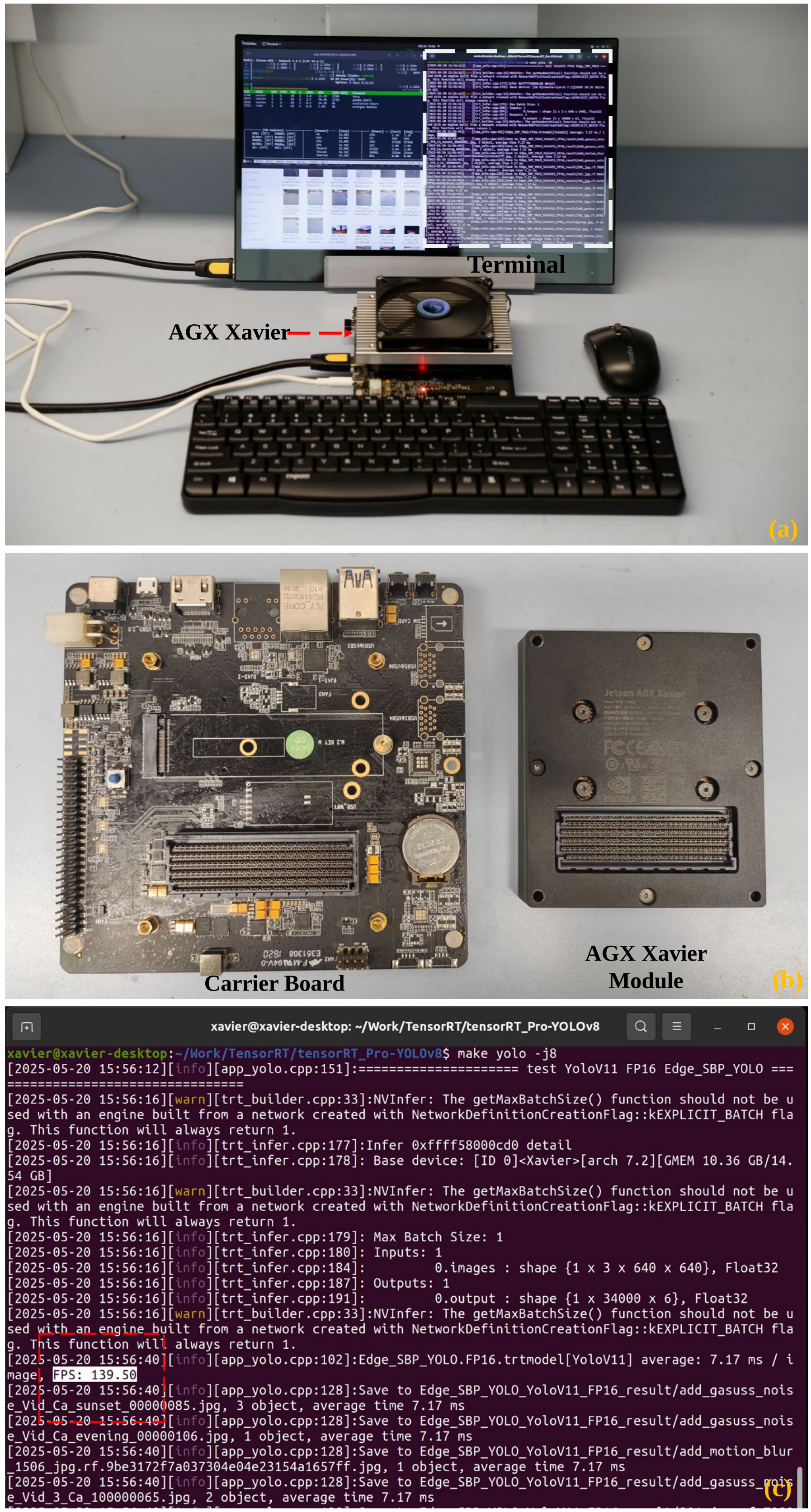


Fig. 10 Experimental hardware setup: **a** experimental platform; **b** Jetson AGX Xavier-based hardware configuration; **c** display terminal for real-time inference results

Table 8 Hardware-level ablation study on NVIDIA Jetson Platform

Model	FPS	Lat. (ms)	Power (W)	E/frame (mJ)	Eff. (FPS/W)	GPU (%)	RAM (%)	Quant. size (MB)
YOLOv11n (INT8)	229.83	4.35	20.49	89.15	11.22	91.19	28.38	4.79
YOLOv11n (FP16)	186.12	5.37	22.61	121.48	8.23	91.45	29.97	6.73
YOLOv11n (FP32)	105.51	9.48	26.88	254.76	3.93	95.23	30.01	13.27
+P2 (INT8)	152.78	6.55	23.36	152.89	6.54	93.49	30.06	5.53
+P2 (FP16)	124.03	8.06	25.46	205.27	4.87	94.07	30.11	7.42
+P2 (FP32)	70.58	14.17	28.81	408.19	2.45	96.10	30.19	14.35
+P2+LEDH (INT8)	169.12	5.91	23.75	140.43	7.12	93.10	30.18	5.04
+P2+LEDH (FP16)	141.80	7.05	24.85	175.25	5.71	93.73	30.17	6.61
+P2+LEDH (FP32)	82.10	12.18	28.14	342.75	2.92	95.85	30.18	11.76
+P2+LEDH+Ghost (INT8)	168.31	5.94	23.48	139.50	7.17	93.14	30.18	4.76
+P2+LEDH+Ghost (FP16)	141.28	7.08	24.68	174.72	5.73	93.78	30.19	6.76
+P2+LEDH+Ghost (FP32)	83.92	11.92	28.19	335.92	2.98	95.68	30.19	10.71
SBP-YOLO (INT8)	155.78	6.42	23.02	147.74	6.77	90.01	30.40	5.24
SBP-YOLO (FP16)	139.50	7.17	24.39	174.84	5.72	91.07	30.41	6.03
SBP-YOLO (FP32)	84.52	11.83	27.72	327.97	3.05	94.23	30.47	9.60

Models are incrementally augmented in order (Ghost = GhostConv)

Lat. latency, *E/F* energy per frame, *Eff.* FPS per watt, *GPU/RAM* average utilization

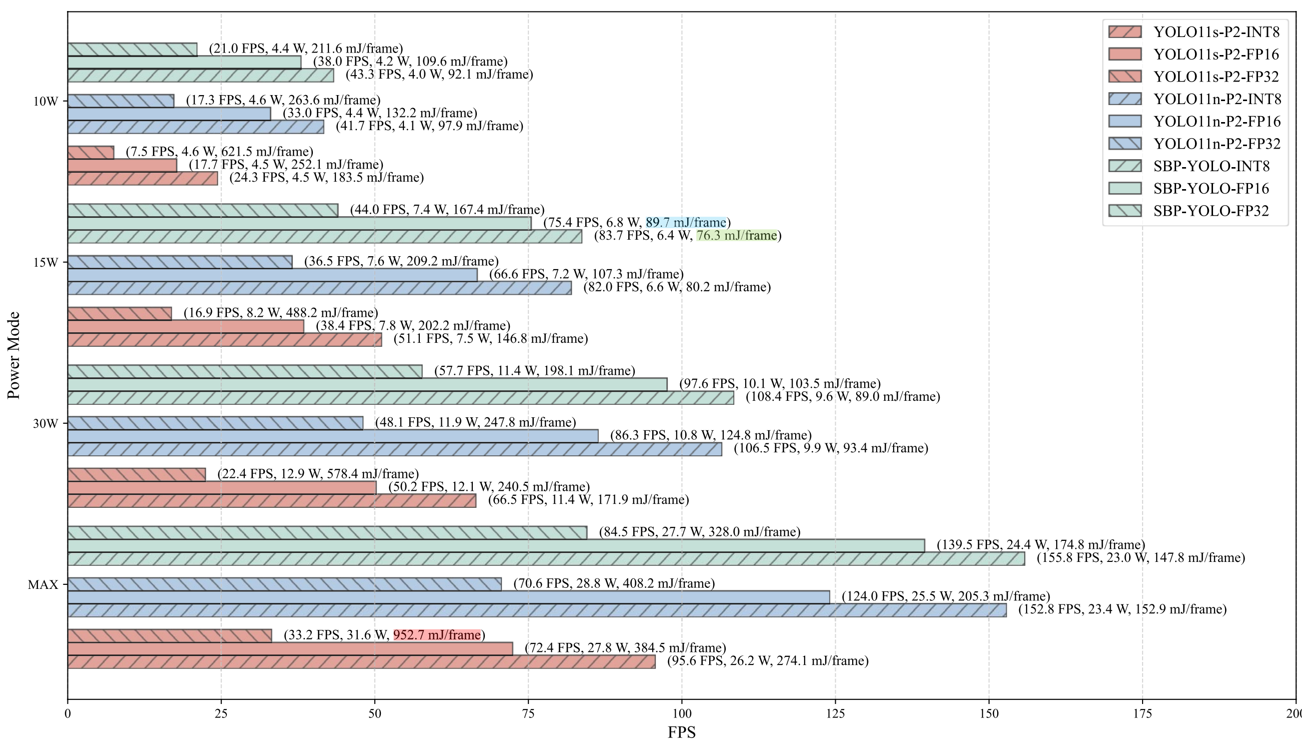


Fig. 11 Comparison of inference FPS and energy consumption per frame for YOLO11-P2, YOLO11-P2s, and SBP-YOLO under varying power configurations

Fig. 12 Comparison of recognition performance of different TensorRT quantization methods

Actuator response times vary significantly across suspension technologies. Magnetorheological dampers (MRD) and continuously variable damping (CDC) actuators typically respond within 15–25 ms [59–61], while air-spring systems exhibit slower pressure-regulation dynamics due to gas compressibility and flow limitations [62]. Advanced control strategies including model predictive and preview control contribute approximately 50 ms to the computational overhead in t_{ctrl} [9].

As detailed in Table 9, the proposed SBP-YOLO achieves a perception latency of 7.17 ms (139.5 FPS) under TensorRT FP16 optimization. At the total latency budget of 77.17 ms, this accounts for only 9.3% of the available time, leaving substantial temporal margin for controller computation and actuator response. These results demonstrate the framework's capability to support real-time suspension preview control across typical operating scenarios, covering vehicle speeds of 10–30 m/s (36–108 km/h) and preview distances of 5–20 m, while accommodating varying system requirements.

7 Conclusion

This paper presents SBP-YOLO, a lightweight framework for real-time road anomaly detection to support predictive suspension control. Based on YOLOv11, the architecture incorporates a P2 detection head with LEDH using shared GroupConv to enhance small-target sensitivity while reducing computational cost. The integration of GhostConv and VoVGSCSPC modules further improves multi-scale feature extraction. Through a hybrid training approach combining NWD loss, knowledge distillation, and advanced augmentation, the framework achieves a 5.8% mAP improvement over YOLOv11n. After TensorRT FP16 optimization, it attains 139.5 FPS on the Jetson AGX Xavier platform, achieving a 12.5% speed improvement compared to the P2-enhanced YOLOv11n. SBP-YOLO also shows significant energy efficiency, consuming 174.84 mJ per frame (14.8% reduction vs YOLOv11-P2). Under 15W power constraints with INT8 quantization, energy consumption decreases further to 76.3 mJ per frame. These results confirm the framework's suitability for embedded

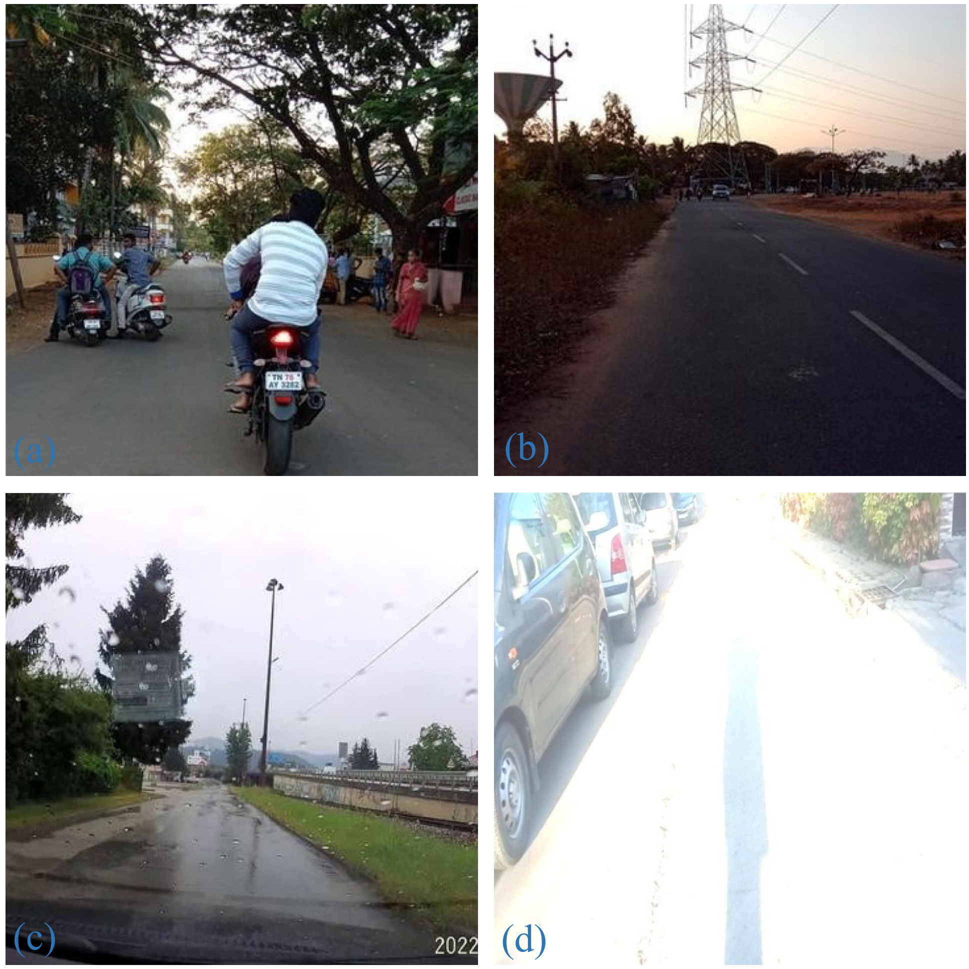


Fig. 13 Characteristic failure cases: **a** occluded speed bumps, **b** low-texture distant targets, **c** rain-obscured road defects, **d** overexposure artifacts

Table 9 Preview time margins ($t_{total} \approx 77.17$ ms)

v (m/s)	D (m)	$T_{preview}$ (ms)	Margin (ms)
10	5	500	422.83
20	5	250	172.83
30	5	167	89.83
20	10	500	422.83
30	10	333	255.83
30	20	667	589.83

deployment in real-time suspension systems. Future work will explore infrared integration and multimodal detection for enhanced robustness in diverse environments.

Author contributions Chuanqi Liang contributed to conceptualization, methodology, software development, model implementation, validation, and original draft writing. Miao Yu contributed to investigation, data preparation.Jie Fu contributed to data labeling. Jie Yuan, Linlin Gou, and Lei Luo contributed to review and editing. All authors have read and approved the final manuscript.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Li, W., Du, X., Xia, D., Fu, J., Yu, M.: Robust h_∞ control for magnetorheological suspension system of all-terrain vehicles with parameter uncertainties and time delay. Veh. Syst. Dyn. 1–26 (2025). https://doi.org/10.1080/00423114.2025.2501993
2. Yu, M., Xia, D., Li, W., Han, G., Du, X., Fu, J.: Research on finite frequency robust h_∞ control of all-terrain vehicle magnetorheological suspension system. Mech. Syst. Signal Process. **234**, 112803 (2025)

3. Yu, M., Evangelou, S.A., Dini, D.: Advances in active suspension systems for road vehicles. Engineering **33**, 160–177 (2024)
4. Ferhath, A.A., Kasi, K.: The evolution of damper technology for enhanced ride comfort and vehicle handling in vehicle suspension system. Int. J. Dyn. Control **12**(11), 3908–3946 (2024)
5. Wambold, J.C., Zimmer, R.A., Ivey, D., Sicking, D.L.: Roughness, holes, and bumps. In: Roadway Surface Discontinuities on Safety, vol. 11 (2009)
6. Yin, Y., Fu, W., Ma, X., Yu, J., Li, X., Dong, Z.: Road surface pits and speed bumps recognition based on acceleration sensor. IEEE Sens. J. **24**(7), 10 669-10 679 (2024)
7. Aguilar-González, A., Medina Santiago, A.: Road event detection and classification algorithm using vibration and acceleration data. Algorithms **18**(3), 127 (2025)
8. Achnib, A., Sename, O.: Discrete-time multi-model preview control: application to a real semi-active automotive suspension system. Control. Eng. Pract. **137**, 105553 (2023)
9. Li, W., Liang, H., Xia, D., Fu, J., Yu, M.: Explicit model predictive control of magnetorheological suspension for all-terrain vehicles with road preview. Smart Mater. Struct. **33**(3), 035037 (2024)
10. Jung, J.Y., Lee, C.: Preview model predictive control of semi-active suspension for speed bump. Int. J. Automot. Technol. **26**(4), 1115–1126 (2025)
11. Asad, M.H., Khaliq, S., Yousaf, M.H., Ullah, M.O., Ahmad, A.: Pothole detection using deep learning: a real-time and AI-on-the-edge perspective. Adv. Civ. Eng. **2022**(1), 9221211 (2022)
12. Bhavana, N., Kodabagi, M.M., Kumar, B.M., Ajay, P., Muthukumaran, N., Ahilan, A.: POT-YOLO: real-time road potholes detection using edge segmentation-based Yolo v8 network. IEEE Sens. J. **24**(15), 24 802-24 809 (2024)
13. Liu, S., Zha, J., Sun, J., Li, Z., Wang, G.: EdgeYOLO: an edge-real-time object detector. In: 42nd Chinese Control Conference (CCC), vol. 2023, pp. 7507–7512. IEEE (2023)
14. Zhao, J., Liu, S., Tao, H., Liu, W.: Slim-YOLOv8: a fast and accurate algorithm for surface defect detection of steel strips. Ironmak. Steelmak. (2024). https://doi.org/10.1177/03019233241266717
15. Bučko, B., Lieskovská, E., Zábovská, K., Zábovskỳ, M.: Computer vision based pothole detection under challenging conditions. Sensors **22**(22), 8878 (2022)
16. Youwai, S., Chaiyaphat, A., Chaipetch, P.: YOLO9tr: a lightweight model for pavement damage detection utilizing a generalized efficient layer aggregation network and attention mechanism. J. Real-Time Image Proc. **21**(5), 163 (2024)
17. Lyu, R.: NanoDet-Plus: super fast and high accuracy lightweight anchor-free object detection model. https://github.com/RangiLyu/nanodet (2021)
18. Girshick, R., Donahue, J., Darrell, T., Malik, J.: Rich feature hierarchies for accurate object detection and semantic segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 580–587 (2014)
19. Girshick, R.: Fast R-CNN. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 1440–1448 (2015)
20. Ren, S., He, K., Girshick, R., Sun, J.: Faster R-CNN: towards real-time object detection with region proposal networks. IEEE Trans. Pattern Anal. Mach. Intell. **39**(6), 1137–1149 (2016)
21. Redmon, J., Farhadi, A.: YOLOv3: an incremental improvement. arXiv preprint arXiv:1804.02767 (2018)
22. Jocher, G., Stoken, A., Borovec, J., Changyu, L., Hogan, A., Diaconu, L., Poznanski, J., Yu, L., Rai, P., Ferriday, R., et al.: Ultralytics/YOLOv5: v3. 0. Zenodo (2020)
23. Li, C., Li, L., Jiang, H., Weng, K., Geng, Y., Li, L., Ke, Z., Li, Q., Cheng, M., Nie, W., et al.: YOLOv6: a single-stage object detection framework for industrial applications. arXiv preprint arXiv:2209.02976 (2022)
24. Sohan, M., Sai Ram, T., Rami Reddy, C.V.: A review on yolov8 and its advancements. In: International Conference on Data Intelligence and Cognitive Informatics, pp. 529–545. Springer (2024)
25. Wang, C.-Y., Yeh, I.-H., Mark Liao, H.-Y.: YOLOv9: learning what you want to learn using programmable gradient information. In: European Conference on Computer Vision, pp. 1–21. Springer (2024)
26. Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., et al.: YOLOv10: real-time end-to-end object detection. Adv. Neural. Inf. Process. Syst. **37**, 107984–108011 (2024)
27. Khanam, R., Hussain, M.: YOLOv11: an overview of the key architectural enhancements. arXiv preprint arXiv:2410.17725 (2024)
28. Chileshe, M., Nyirenda, M., Kaoma, J.: Early detection of sexually transmitted infections using YOLO 12: a deep learning approach. Open J. Appl. Sci. **15**(4), 1126–1144 (2025)
29. Lei, M., Li, S., Wu, Y., Hu, H., Zhou, Y., Zheng, X., Ding, G., Du, S., Wu, Z., Gao, Y.: YOLOv13: real-time object detection with hypergraph-enhanced adaptive visual perception. arXiv preprint arXiv:2506.17733 (2025)
30. Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., Zagoruyko, S.: End-to-end object detection with transformers. In: European Conference on Computer Vision, pp. 213–229. Springer (2020)
31. Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y., Chen, J.: DETRs beat YOLOs on real-time object detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 16965–16974 (2024)
32. Feng, Y., Huang, J., Du, S., Ying, S., Yong, J.-H., Li, Y., Ding, G., Ji, R., Gao, Y.: Hyper-YOLO: when visual object detection meets hypergraph computation. IEEE Trans. Pattern Anal. Mach. Intell. **47**, 2388–2401 (2024)
33. Wang, Z., Li, C., Xu, H., Zhu, X.: Mamba YOLO: SSMs-based YOLO for object detection. arXiv e-prints, pp. arXiv-2406 (2024)
34. Park, S.-S., Tran, V.-T., Lee, D.-E.: Application of various yolo models for computer vision-based real-time pothole detection. Appl. Sci. **11**(23), 11229 (2021)
35. Wang, R., Luo, X., Ye, Q., Jiang, Y., Liu, W.: Research on visual perception of speed bumps for intelligent connected vehicles based on lightweight FPNet. Sensors **24**(7), 2130 (2024)
36. Khan, M., Raza, M.A., Abbas, G., Othmen, S., Yousef, A., Jumani, T.A.: Pothole detection for autonomous vehicles using deep learning: a robust and efficient solution. Front. Built Environ. **9**, 1323792 (2024)
37. Dewangan, D.K., Sahu, S.P., Arya, K.V.: Vision-sensor enabled multilayer CNN scheme and impact analysis of learning rate parameter for speed bump detection in autonomous vehicle system. IEEE Sens. Lett. **8**(3), 1–4 (2024)
38. Peralta-López, J.-E., Morales-Viscaya, J.-A., Lázaro-Mata, D., Villaseñor-Aguilar, M.-J., Prado-Olivarez, J., Pérez-Pinal, F.-J., Padilla-Medina, J.-A., Martínez-Nolasco, J.-J., Barranco-Gutiérrez, A.-I.: Speed bump and pothole detection using deep neural network with images captured through zed camera. Appl. Sci. **13**(14), 8349 (2023)
39. Chandak,P., Chaurasia, S., Patil, M.: Identification of potholes and speed bumps using SSD and YOLO. In: International Conference on Intelligent Computing and Networking, pp. 1–20. Springer (2024)
40. Safyari, Y., Mahdianpari, M., Shiri, H.: A review of vision-based pothole detection methods using computer vision and machine learning. Sensors **24**(17), 5652 (2024)
41. Shen, Y., Jing, K., Sun, K., Liu, C., Yang, Y., Liu, Y.: Review of uneven road surface information perception methods for suspension preview control. Sensors **25**(18), 5884 (2025)
42. Shen, J., Zhou, W., Liu, N., Sun, H., Li, D., Zhang, Y.: An anchor-free lightweight deep convolutional network for vehicle

detection in aerial images. IEEE Trans. Intell. Transp. Syst. **23**(12), 24 330-24 342 (2022)
43. Shen, J., Liu, N., Sun, H., Li, D., Zhang, Y.: An instrument indication acquisition algorithm based on lightweight deep convolutional neural network and hybrid attention fine-grained features. IEEE Trans. Instrum. Meas. **73**, 1–16 (2024)
44. Shen, J., Liu, N., Sun, H., Li, D., Zhang, Y., Han, L.: An algorithm based on lightweight semantic features for ancient mural element object detection. NPJ Herit. Sci. **13**(1), 70 (2025)
45. Shen, J., Liu, N., Xu, C., Sun, H., Xiao, Y., Li, D., Zhang, Y.: Finger vein recognition algorithm based on lightweight deep convolutional neural network. IEEE Trans. Instrum. Meas. **71**, 1–13 (2021)
46. Han, K., Wang, Y., Tian, Q., Guo, J., Xu, C., Xu, C.: GhostNet: more features from cheap operations. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 1580–1589 (2020)
47. Li, H., Li, J., Wei, H., Liu, Z., Zhan, Z., Ren, Q.: Slim-neck by GSConv: a lightweight-design for real-time detector architectures. J. Real-Time Image Proc. **21**(3), 62 (2024)
48. Wang, J., Xu, C., Yang, W., Yu, L.: A normalized Gaussian Wasserstein distance for tiny object detection. arXiv preprint arXiv:2110.13389 (2021)
49. Nienaber, S., Booysen, M., Kroon, R.: Detecting potholes using simple image processing techniques and real-world footage. In: Southern African Transport Conference (2015)
50. Varma, V., Adarsh, S., Ramachandran, K., Nair, B.B.: Real time detection of speed hump/bump and distance estimation with deep learning using GPU and zed stereo camera. Procedia Comput. Sci. **143**, 988–997 (2018)
51. Buslaev, A., Iglovikov, V.I., Khvedchenya, E., Parinov, A., Druzhinin, M., Kalinin, A.A.: Albumentations: fast and flexible image augmentations. Information **11**(2), 125 (2020)
52. Wang, Q., Liu, L., Yu, W., Chen, S., Gong, J., Chen, P.: BCKD: block-correlation knowledge distillation. In: 2023 IEEE International Conference on Image Processing (ICIP), pp. 3225–3229. IEEE (2023)
53. Shu, C., Liu, Y., Gao, J., Yan, Z., Shen, C.: Channel-wise knowledge distillation for dense prediction. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 5311–5320 (2021)
54. Peng, G., Wang, K., Ma, J., Cui, B., Wang, D.: AGRI-YOLO: a lightweight model for corn weed detection with enhanced YOLO v11n. Agriculture **15**(18), 1971 (2025)
55. Zheng, H., Liu, C., Zhong, L., Wang, J., Huang, J., Lin, F., Ma, X., Tan, S.: An android-smartphone application for rice panicle detection and rice growth stage recognition using a lightweight YOLO network. Front. Plant Sci. **16**, 1561632 (2025)
56. Zhang, S., Jin, X., Lou, Z., Wang, S., Lu, S., He, Y.: LSDF-Net: an efficient lightweight defect detection method for ultrasonic welding surfaces. J. Adv. Join. Process. **12**, 100339 (2025). https://doi.org/10.1016/j.jajp.2025.100339
57. Wang, X., Wan, S., Li, Z., Chen, X., Zhang, B., Wang, Y.: ECL-TEAR: lightweight detection method for multiple types of belt tears. Measurement **251**, 117269 (2025)
58. Zhou, M.: TensorRT Pro-YOLOv8. https://github.com/Melody-Zhou/tensorRT_Pro-YOLOv8. Accessed 31 July 2025
59. Liang, H., Fu, J., Li, W., Xia, D., Luo, L., Yu, M.: Structural design and multi-objective optimization of a novel asymmetric magnetorheological damper. Smart Mater. Struct. **33**(3), 035041 (2024)
60. Li, W., Liang, H., Xia, D., Fu, J., Luo, L., Yu, M.: Research on current drive system of magnetorheological damper based on fuzzy PI control. Materials **15**(24), 8893 (2022)
61. Ji, G., Zhang, L., Cai, M., Meng, X., Du, Z., Ruan, J., Guan, S., Liu, Z.: Research on variable universe fuzzy PID control for semi-active suspension with CDC dampers based on dynamic adjustment functions. Sci. Rep. **14**(1), 3442 (2024)
62. Yin, C., Zhao, D., Zhang, J., Wang, S., Xu, X., Sun, X., Shi, D.: Body height robust control of automotive air suspension system using finite-time approach. Asian J. Control **24**(2), 859–871 (2022)